cheetahs. Similarly, if space and time turn out to be composite entities, it wouldn't mean that their familiar manifestations, from Newton's bucket to Einstein's gravity, are illusory; there is little doubt that space and time will retain their all-embracing positions in experiential reality, regardless of future developments in our understanding. Instead, composite spacetime would mean that an even more elemental description of the universe—one that is spaceless and timeless—has yet to be discovered. The illusion, then, would be one of our own making: the erroneous belief that the deepest understanding of the cosmos would bring space and time into the sharpest possible focus. Just as the hardness of a cannonball, the smell of the rose, and the speed of the cheetah disappear when you examine matter at the atomic and subatomic level, space and time may similarly dissolve when scrutinized with the most fundamental formulation of nature's laws.

That spacetime may not be among the fundamental cosmic ingredients may strike you as somewhat far-fetched. And you may well be right. But rumors of spacetime's impending departure from deep physical law are not born of zany theorizing. Instead, this idea is strongly suggested by a number of well-reasoned considerations. Let's take a look at some of the most prominent.

Quantum Averaging

In Chapter 12 we discussed how the fabric of space, much like everything else in our quantum universe, is subject to the jitters of quantum uncertainty. It is these fluctuations, you'll recall, that run roughshod over point-particle theories, preventing them from providing a sensible quantum theory of gravity. By replacing point particles with loops and snippets, string theory spreads out the fluctuations—substantially reducing their magnitude—and this is how it yields a successful unification of quantum mechanics and general relativity. Nevertheless, the diminished spacetime fluctuations certainly still exist (as illustrated in the next-to-last level of magnification in Figure 12.2), and within them we can find important clues regarding the fate of spacetime.